

Sarah K. Swiston, Ph.D.

Postdoctoral Teaching Fellow

Morningside University — Sioux City, Iowa

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I am a postdoctoral instructor and researcher at Morningside University. My work centers on teaching project-based courses for first-year STEM students and studying the impact of these courses on student agency, scientific identity, and scientific reasoning. I am also involved in continuing biological research focused on developing mathematical models to investigate how species evolve over time and move through space.

EMPLOYMENT

Morningside University Sioux City, IA Postdoctoral teaching fellow	2026 – Present
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EDUCATION

Washington University Saint Louis, MO Ph.D. in Ecology & Evolutionary Biology	2020 – 2026
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University of Oklahoma Norman, OK B.A. in Biology, B.S. in Mathematics, minor in Music, Summa Cum Laude	2016 – 2020
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FELLOWSHIPS & AWARDS

National Science Foundation Graduate Research Fellowship Program	2022 – 2026
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Washington University Center for Teaching and Learning Educational Development Graduate Fellowship	2025
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SSB Publisher's Award for Excellence in Systematic Research

ASN Outstanding Student Poster Award

Morris B. Rettner Scholarship

Carl & Pat Bynum Riggs Scholarship	2018
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National Merit Scholar	2016
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Illinois State Scholar

RESEARCH EXPERIENCE

Postdoctoral Teaching Fellow Dr. Brian McFarland at Morningside University	2026 – Present
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Ph.D. Student Dr. Michael Landis at Washington University	2021 – 2026
• Biogeography & phylogenetic comparative methods • R package & Docker image development • Biogeography of <i>Liolaemus</i> lizards, Australian Sphenomorphine lizards, and Hawaiian plants • Running computational workshops and assessing their efficacy for motivating researchers to use new methods	
Rotation Student Dr. Jonathan Losos at Washington University	2020 – 2021
• Global distribution of arboreal lizards • Herpetology field experience	
Research Student & Technician Dr. Katharine Marske at University of Oklahoma	2019 – 2020
• Lab work including beetle DNA extraction, PCR, and gel electrophoresis • Data analysis in Geneious, QGIS, BEAST, R	
Research Apprentice University of Oklahoma Biological Station	2018
• Ant behavioral studies • Ant heat tolerance testing • Entomology field experience	
Lab Volunteer Sam Noble Oklahoma Museum of Natural History: Herpetology	2017 – 2018
• Herpetology field experience • Specimen preparation and maintenance • Tissue sampling • Lab work including DNA extraction and qPCR	

PUBLICATIONS

Swiston, S. K., & Landis, M. J. (2025). Testing relationships between multiple regional features and biogeographic processes of speciation, extinction, and dispersal. *Systematic Biology*, 74(2), 282–300.

PREPRINTS & IN PREP

Lichter-Marck, I., **Swiston, S. K.**, Mendes, F. K., May, M., Neupane, S., Rønsted, N., Baldwin, B., Wagner, W., Wood, K., Zapata, F., Landis, M. J. (2025). The Goldilocks effect drives plant diversification on middle-aged Hawaiian islands. *bioRxiv*.

Swiston, S. K., Kuehne, L., Moore, R., & Landis, M. J. (2025). Assessing the efficacy of computational workshops and participatory live coding in evolutionary biology. *bioRxiv*.

Swiston, S. K., McHugh, S., & Landis, M. J. (2025). EMPIRE: The ellipse model for phylogenetic inference of range evolution. *bioRxiv*. *In review at: Systematic Biology*

PRESENTATIONS

CIRTL General Meeting **2025**

IGNITE: The Interdisciplinary Graduate Network for Inquiry, Teaching, & Engagement

Co-Presenters: Nicole Strombom

Evolution Conference

EMPIRE: The ellipse model for phylogenetic inference of range evolution

Midwest Ecology and Evolution Conference

Modeling range evolution using ellipses in continuous space

The CIRTl Network Teaching-as-Research Presentations

Assessing the efficacy of computational workshops and participatory live coding in evolutionary biology

Graduate Research Symposium (Washington University)

Assessing the efficacy of computational workshops and participatory live coding in evolutionary biology

Ecology and Evolution Seminar (Washington University)

Designing and disseminating phylogenetic models of historical biogeography

2024

Evolution Conference

A model of range evolution using ellipses in continuous space

Society for the Study of Amphibians and Reptiles Conference

Testing relationships between regional features and biogeographic processes of speciation, extinction, and dispersal in *Liolaemus* lizards

Evolution Conference

Testing relationships between multiple regional features and biogeographic process of speciation, extinction, and dispersal

2023

Living Earth Collaborative Seminar (Washington University)

How the environment shapes diversification and dispersal: A feature-informed phylogenetic method of historical biogeography (+ Lizards!)

International Biogeography Society Conference

Inferring evolutionary rates in a geographic context: Incorporating region features

2022

Midwest Ecology and Evolution Conference

A phylogenetic model of historical biogeography incorporating regional features

Evolution Conference

Why are arboreal lizards a tropical phenomenon?

2021

POSTERS

Evolution Conference

EMPIRE: The ellipse model for phylogenetic inference of range evolution

2025

Graduate Research Symposium (Washington University)

Education in phylogenetic modeling: The pedagogy of tutorials, software, and workshops

Society for the Advancement of Biology Education Research Midwest Conference

Education in phylogenetic modeling: the pedagogy of tutorials, software, and workshops

2024

Evolution Conference

Education in phylogenetic modeling: The pedagogy of tutorials, software, and workshops

University of Oklahoma Undergraduate Research Day

Amphibious infectious disease in Oklahoma: A citizen science project

2017

THESES

Honors Thesis | University of Oklahoma

Phylogeography of New Zealand weevil *Geochus tibialis* (Curculionidae)

2019

PRESS

News Article | Neighbors of Glen Ellyn
A Science Grad Student's Journey

2024

MENTORSHIP

Research Mentor | Washington University

2025 – Present

Mentee: Matthew Pottinger, Washington University in Saint Louis (Undergraduate)

Mentorship:

- Assisted in the development of hypotheses and targets for data collection
- Guided the design of statistical analyses using R
- Workshopped written products

Mentee accomplishments:

- Wrote an undergraduate thesis: Phylogenetic tests of adaptive and nonadaptive hypotheses for same-sex sexual behavior in mammals
- Gave a poster presentation
- Won the 2026 Ralph S. Quatrano prize
- Gave an award talk
- Currently working toward a manuscript for publication

Copy Editor & Reviewer | Journal of Emerging Investigators

2022 – 2026

Research Mentor | Young Scientist Program Summer Focus Program

2022

Mentee: Benjamin Gondzur, Metro Academic and Classical High School (Senior)

Mentorship:

- Designed structured, scaffolded research project using R Markdown
- Activities included readings, coding exercises, and data analysis

Mentee accomplishments:

- Wrote a high school thesis
- Gave an oral presentation

TEACHING PROFESSIONAL DEVELOPMENT

Educational Development Graduate Fellowship | Center for Teaching and Learning

2025 – 2026

- Facilitated graduate student learning communities (EPIC, IGNITE)
- Contributed to workshop development

Member | IGNITE Learning Community

2024 – 2026

(Interdisciplinary Graduate Network for Inquiry, Teaching, and Engagement)

Member | Washington University Education Research Group

2024 – 2026

Teaching Citation | Washington University

2023

Professional Development in Teaching Program

Member | Center for Teaching and Learning Graduate Student Advisory Council

2022 – 2024

Member | EPIC Learning Community

2020 – 2021

(Exploring Practices in the Classroom)

Member | Washington University in Saint Louis Future Educators

2020

Competent in the following tools: Canvas, Zoom, Microsoft Office, Google Workspace

WORKSHOPS ATTENDED

Using Game-Inspired Design to Spark Engagement and Deep Learning Washington University Instructor: Dr. Tori Mondelli, University of Missouri	2025
Sparkling Joy in Biology Education SABER 2025 Conference	
Scholarship of Teaching and Learning Seminar Washington University Center for Teaching and Learning	2024
Introduction to CODON Learning SABER Midwest 2024 Conference	
Data Management Washington University Libraries Data Services Responsible data management, storage, security, and availability	2023
The Inclusive STEM Teaching Project	
Evolutionary Quantitative Genetics Dr. Stevan J. Arnold & Dr. Joe Felsenstein Lecturers: Dr. Patrick Carter, Dr. Adam Jones, Dr. Brian O'Meara, Dr. Samantha Price, Dr. Josef Uyeda	2022
Short Teaching Workshops Washington University Center for Teaching and Learning Applying Cognitive Science (2021), Engaging in Equity Pedagogy (2021), Introduction to Pedagogical Scholarship (2021), Developing Effective Summative Assessments (2022), Planning and Organizing a Class Session (2023), Process Oriented Guided Inquiry Learning – POGIL (2024), Introduction to Team-Based Learning – TBL (2025), Crafting Critical Thought: Hands-On Creativity for Classroom Engagement (2025)	

PROGRAMMING

PhyloDocker Docker Image: hub.docker.com/r/sswiston/phylo_docker	2022 – Present
Competent in the following tools: R, R Devtools, R Shiny, RevBayes, Python, Java, Docker, QGIS, Unix	

GRADUATE COURSEWORK

Population Genetics, Genomics, Community Ecology, Stochastic Processes, Data Structures & Algorithms, Genomics, Ethics and Research Science

REVIEWED FOR

Evolution, Journal of Emerging Investigators, Methods in Ecology & Evolution, Molecular Phylogenetics & Evolution, Proceedings of the Royal Society B, Systematic Biology

SOCIETY MEMBERSHIPS

American Society of Naturalists (ASN), National Association of Biology Teachers (NABT), Partners in Amphibian and Reptile Conservation (PARC), Professional and Organizational Development Network in Higher Education (POD), Society for the Advancement of Biology Education Research (SABER), Society for the Study of Amphibians and Reptiles (SSAR), Society of Systematic Biologists (SSB), Society for the Study of Evolution (SSE)

REFERENCES

Dr. Michael Landis
Assistant Professor, Department of Biology
Washington University in Saint Louis
CB 1137
One Brookings Drive
St. Louis, MO 63130
michael.landis@wustl.edu
314-935-8082
Relationship: Ph.D. Advisor

Dr. Felipe Zapata
Associate Professor, Department of Ecology and Evolutionary Biology
University of California Los Angeles
612 Charles E. Young Drive South
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Los Angeles, CA 90095
fzapata@ucla.edu
Relationship: Collaborator

Dr. Lisa Kuehne
Assistant Director of Graduate and Postdoc Programs
Washington University Center for Teaching and Learning
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Relationship: Teaching Center Fellowship Supervisor